

Decoy feature detection and MS2-linkability of unmatched features

ProteoStar untargeted MS1 feature detector — branch decoy-features 14 July 2026

1. Purpose

Untargeted MS1 feature detection reports far more features than a database search identifies. Many of the extra (“unmatched”) features are real but unidentified; some are noise fit by chance. To characterise that surplus we (a) built three **isotope-envelope decoys** — theoretical envelopes no real peptide can produce — and ran the full detect→refine pipeline with each, so any feature found is found on coincidence; (b) scored every feature with an **envelope-fit cosine** in 2-D (apex) and 3-D ($m/z \times RT$), and measured how well the score separates real features from decoys; and (c) asked, for the surplus of unmatched features that survive a decoy-calibrated score cut, **how many could plausibly be linked to a real MS2 acquisition** in the raw file.

Three benchmark files are used throughout: **10-min** (CA tryptic, Lumos), **65-min** (D. Morgen N-glyco, HF-X), **2-hr** (IonStar spike-in). Target-feature counts and recall reproduce the benchmarking-guide baseline exactly.

2. Methods

2.1 Decoy envelopes

Each decoy replaces the averagine isotope envelope used by the detector’s matched filter, refinement, and score, end to end:

- **shifted spacing** — isotope teeth at 0.94 Da instead of 1.00336 Da (off-lattice).
- **weird averagine** — an exotic Fe+Cl composition (P2 C1 N1 Cl1 Fe1) replacing the peptide backbone; a physically impossible tooth-weight profile.
- **shuffled envelope** — the real averagine tooth weights randomly permuted (on-lattice, wrong ratios).

2.2 Envelope-fit cosine score

For a feature detected with a given envelope, the score is the cosine between the predicted teeth T and the observed peaks O on a shared m/z grid, with an unexplained-peak penalty: $\text{score} = \sum T \cdot O / (\|T\| \cdot \|O\|)$. The **2-D** score uses the apex scan only; the **3-D** score extends the template over an RT window ($\pm 2\sigma$, $\sigma = 0.15$ min) with $T[k, s] = w_k \cdot g(s)$. Every feature — target or decoy — is scored **against the same envelope it was detected with**.

2.3 Target/decoy matching and thresholds

A target feature is **ID-matched** if it matches a reference PSM/quantified peak within 20 ppm in mass and the per-file RT tolerance ($\pm 0.5 / \pm 1.0 / \pm 0.5$ min); otherwise it is **unmatched**. Two operating points on the 2-D score, both calibrated on the strongest decoy (weird averagine):

- **95% recall** — threshold = 5th percentile of ID-matched target scores (keeps 95% of real IDs).
- **95% decoy rejection (“precision”)** — threshold = 95th percentile of the decoy scores (only 5% of decoys pass). This is the cut used for the MS2-linkability analysis.

2.4 MS2 linkability

For every MS2 scan in the raw file we read the isolation window (target m/z and explicit lower/upper bounds) and RT directly from the vendor file (`dump_ms2_precursors`, via `mzdata`). Isolation widths are fixed per method (5.0, 3.4, 2.0 Th for the three files). For each above-threshold feature we build its average isotope ladder at the detected charge and test three nested criteria, all requiring the MS2 to fire **while the feature elutes or immediately after** (RT in [RT start, RT end + 0.2 min]):

1. **window overlaps a charge state** — some MS2 isolation window covers any isolable isotope m/z ($\geq 5\%$);
2. **centre on any isotope** — some window is *centred* (± 20 ppm) on any isotope of the feature;
3. **centre on the most-abundant isotope** — some window is centred (± 20 ppm) on the feature's base isotope (the “bonus” criterion).

The same logic is applied to two controls: **ID-matched targets** (real, were identified — an MS2 must exist) and **weird-average decoys** (fake features, real MS2 list — the coincidental-link background).

3. Score separates targets from decoys

All three decoys separate ID-matched targets from decoys. Figure 1 overlays the ID-matched-target score distribution (filled) and the unmatched-target distribution (outline) against each decoy, per file. Decoys pile up at low cosine; real features at high cosine, with unmatched features spanning both.

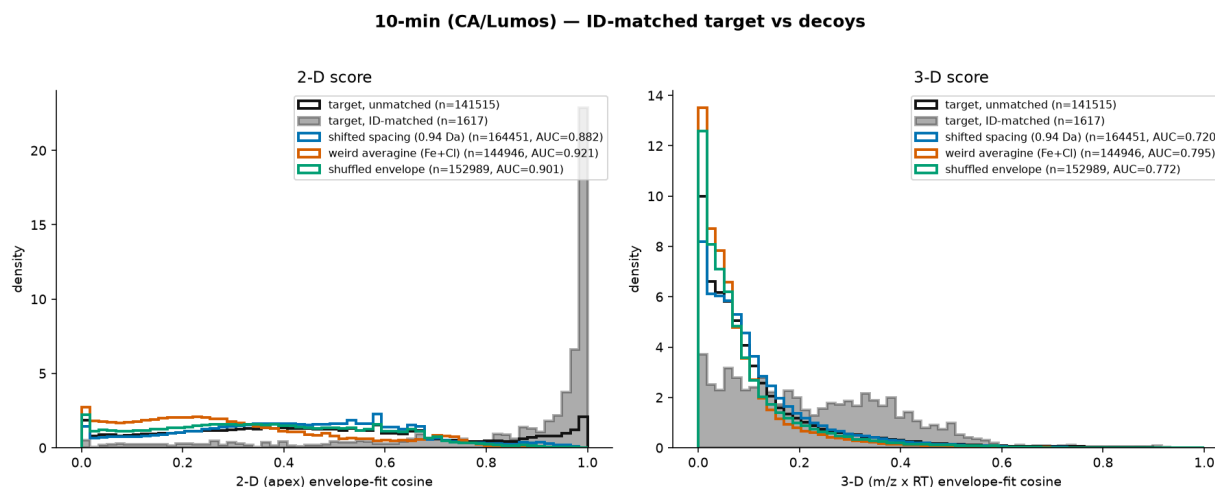
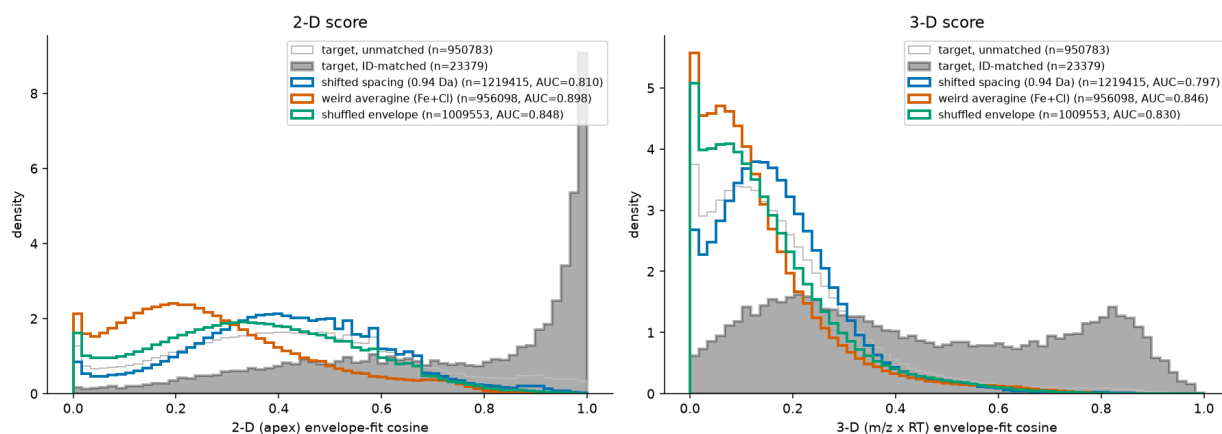
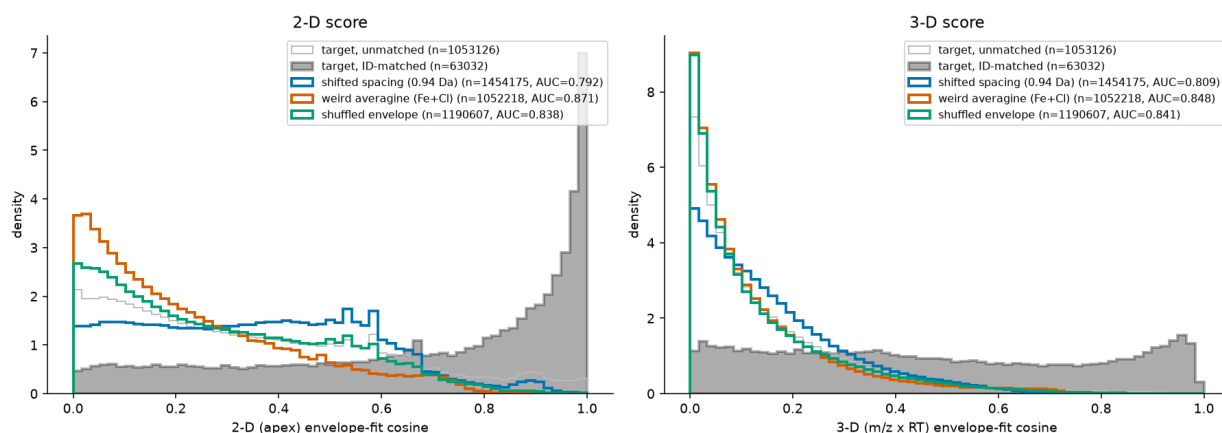


Figure 1: Figure 1 — 2-D and 3-D envelope-fit cosine distributions: ID-matched target vs unmatched target vs decoys, per file (10-min, 65-min, 2-hr top to bottom).

65-min (glyco) — ID-matched target vs decoys



2-hr (IonStar) — ID-matched target vs decoys



ROC and AUC

Figure 2 shows the ROC curves (positive = ID-matched target, negative = decoy). Table 1 lists the AUCs. The **2-D** score is the stronger separator on the fast 10-min gradient; 2-D and 3-D converge, and 3-D edges ahead, on the longer 65-min and 2-hr gradients (more RT sampling to exploit). **Weird averagine** is the hardest decoy to reject and is used to calibrate the thresholds.

10-min (CA/Lumos) — ROC: ID-matched target vs decoy

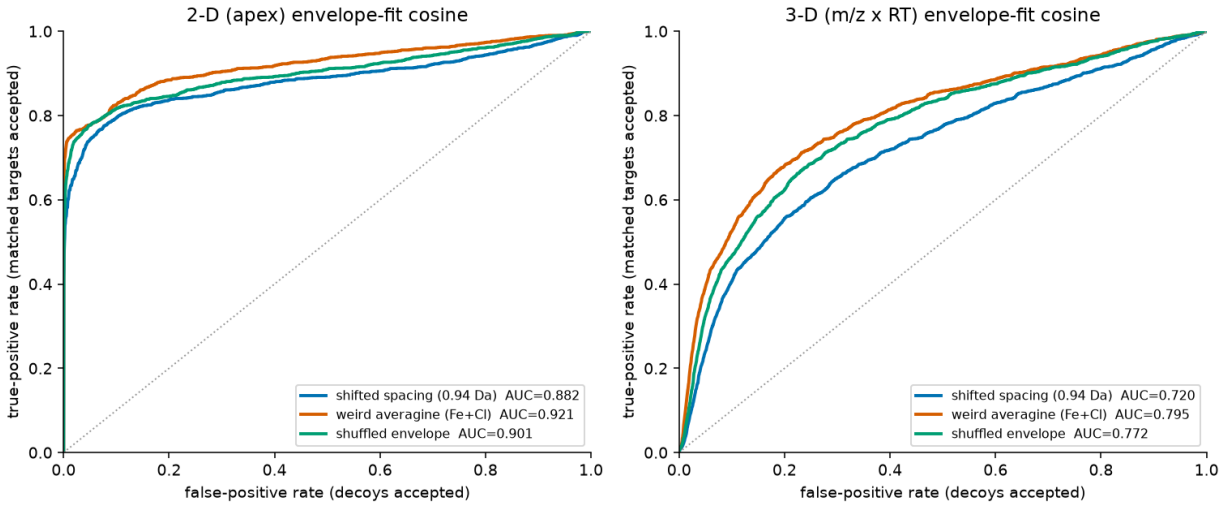
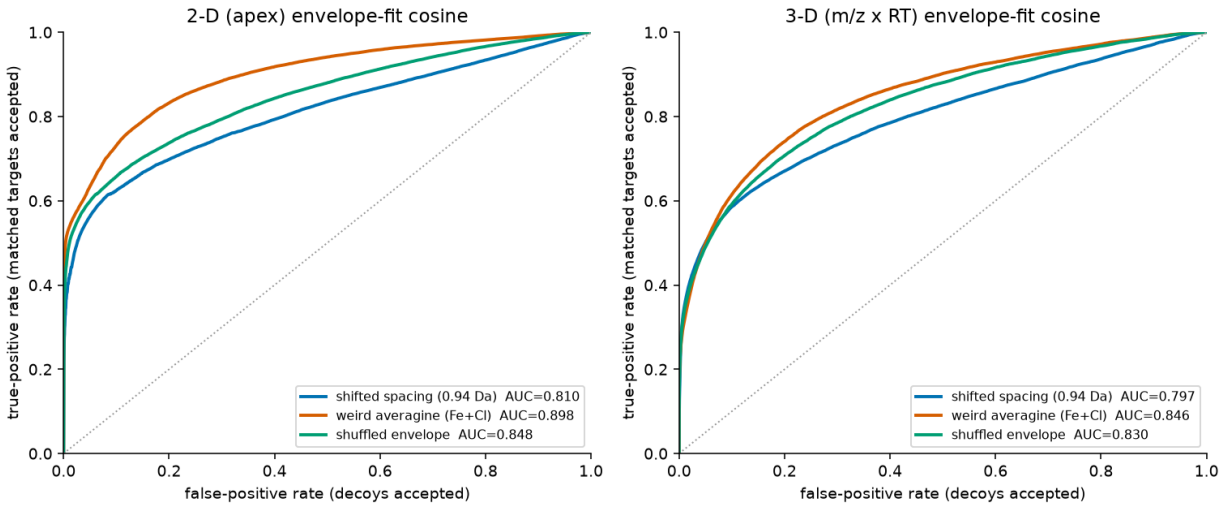


Figure 2: Figure 2 — ROC curves, ID-matched target vs each decoy, 2-D and 3-D score, per file.

65-min (glyco) — ROC: ID-matched target vs decoy



2-hr (IonStar) — ROC: ID-matched target vs decoy

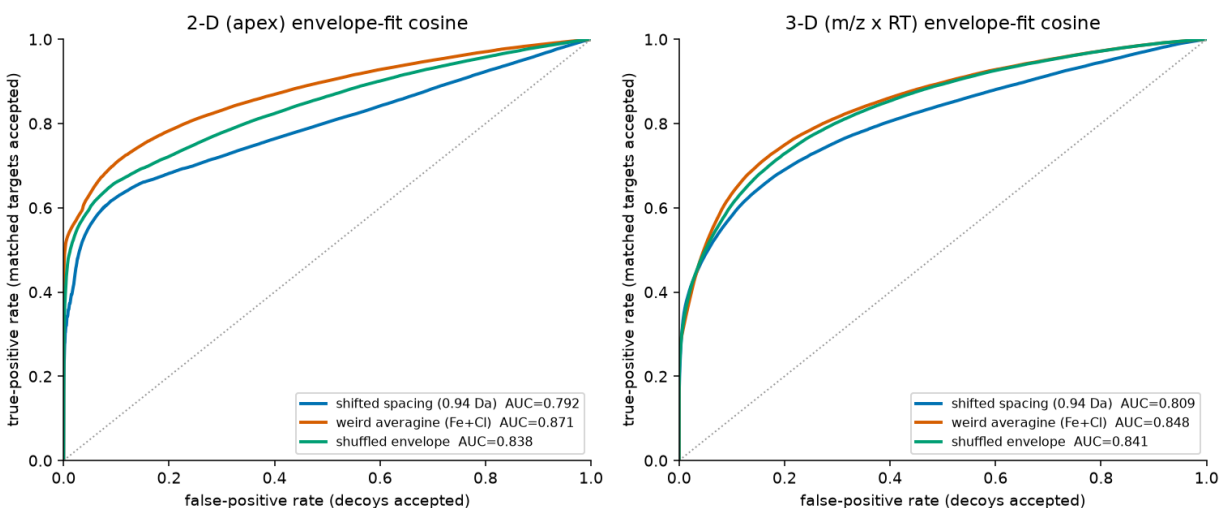


Table 1: **Table 1.** ROC-AUC (ID-matched target vs decoy; 1.0 = perfect separation).

File	shifted 2-D	shifted 3-D	weird 2-D	weird 3-D	shuffled 2-D	shuffled 3-D
10-min	0.882	0.720	0.921	0.795	0.901	0.772
65-min	0.810	0.797	0.898	0.846	0.848	0.830
2-hr	0.792	0.809	0.871	0.848	0.838	0.841

4. Recall / precision operating points

Table 2: **Table 2.** Threshold at **95% recall** of ID-matched targets (weird-averagine calibration), and how much surplus survives it. At this loose cut the score barely filters: most unmatched features and the majority of decoys clear the bar.

File	2-D thr	unmatched $\geq$ thr / total	decoys passing
10-min	0.206	112,953 / 141,515 (80%)	60.4%
65-min	0.222	765,061 / 950,783 (80%)	54.9%
2-hr	0.091	862,525 / 1,053,126 (82%)	69.9%

Table 3: **Table 3.** Threshold that **rejects 95% of decoys** (the cut used below). This sits in the clean upper range of the score and still retains most real features. “Real recall” = fraction of ID-matched targets retained.

File	2-D thr	real recall	unmatched $\geq$ thr / total	decoys passing
10-min	0.715	77.7%	29,000 / 141,515 (20%)	5.0%
65-min	0.644	63.4%	179,522 / 950,783 (19%)	5.0%
2-hr	0.626	63.3%	166,652 / 1,053,126 (16%)	5.0%

The 16–20% of unmatched features that survive the decoy-rejecting cut are the candidate real-but- unidentified set examined next.

5. How many unmatched features are MS2-linkable?

Figure 3 and Table 4 give the three linkability criteria for the above-threshold features, with the matched-target (real) and weird-decoy (null) controls.

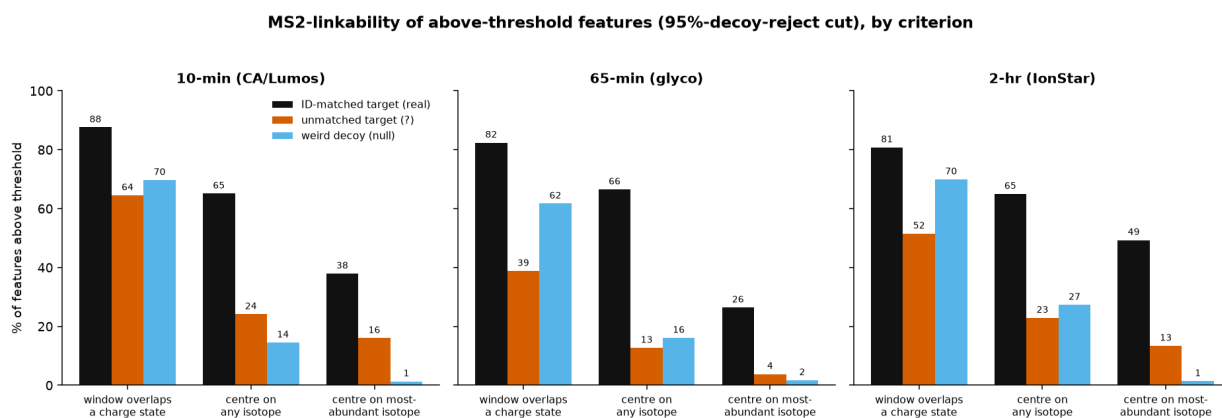


Figure 3: Figure 3 — MS2-linkability rates by criterion. Black = ID-matched targets (real control), orange = unmatched targets (the question), blue = weird decoys (coincidence null).

Table 4: **Table 4.** MS2-linkability of features above the 95%-decoy-reject threshold (% of each set). Sets: *matched* = ID-matched target (real control), *unmatched* = unmatched target (the question), *decoy* = weird-average decoy (coincidence null).

File	set	n	window overlap	centre / any iso	centre / base iso
10-min	matched	1,257	87.6%	65.1%	37.9%
10-min	unmatched	29,000	64.5%	24.1%	16.0%
10-min	decoy	7,248	69.7%	14.5%	1.3%

File	set	n	window overlap	centre / any iso	centre / base iso
65-min	matched	14,828	82.4%	66.5%	26.4%
65-min	unmatched	179,522	38.8%	12.7%	3.8%
65-min	decoy	47,805	61.7%	16.1%	1.7%
2-hr	matched	39,927	80.8%	65.0%	49.3%
2-hr	unmatched	166,652	51.5%	22.9%	13.4%
2-hr	decoy	52,615	69.9%	27.3%	1.4%

The loose criterion is coincidence-dominated. “Window overlaps a charge state during elution” is satisfied by 62–70% of the *decoys* — as often as, or more often than, the unmatched targets — because the wide (2–5 Th) isolation windows and dense MS2 sampling cover almost any *m/z* that is present during a peak’s elution. This criterion therefore carries essentially no evidence of genuine targeting.

The exact-centre criterion is the clean discriminator. Requiring the isolation window to be *centred* on the feature’s most-abundant isotope drops the decoy background to **1.3–1.7%**, while real ID-matched features score 26–49% and the unmatched surplus scores 3.8–16.0% — well above the null. Subtracting the decoy coincidence background gives the number of unmatched features whose MS2 selection cannot be explained by chance:

Table 5: **Table 5.** Unmatched above-threshold features plausibly linked to a genuine MS2 acquisition (isolation centre on the base isotope, during elution), after subtracting the decoy coincidence rate.

File	unmatched $\geq$ thr	centre-hit	coincidence null	net linked	enrichment vs null
10-min	29,000	4,641 (16.0%)	1.3%	$\approx$ 4,300	12.3×
65-min	179,522	6,880 (3.8%)	1.7%	$\approx$ 3,800	2.2×
2-hr	166,652	22,356 (13.4%)	1.4%	$\approx$ 20,100	9.9×

6. Interpretation and caveats

- Roughly **4,300 / 3,800 / 20,100** unmatched features (10-min / 65-min / 2-hr) that clear a decoy-calibrated score cut were also *deliberately selected* for MS2 — the instrument centred an isolation window on their most-abundant isotope while they eluted — at a rate far above chance (12×, 2×, 10× the decoy background). These are strong candidates for real, unidentified species: they look like peptides by envelope shape **and** were independently deemed worth fragmenting by the instrument, yet no database ID was assigned (unsequenced, modified, or search-space limited).
- The unmatched surplus sits **between** the decoy null and the identified-feature control on every criterion — consistent with a mixture of genuine unidentified features and chance detections, not a single population.

- **Do not use the loose “isolation-window overlap” criterion** as evidence of reality: its coincidence background (the decoy null) is 60–70%, so it over-counts massively.
- **65-min undercounts.** It is a glyco dataset: the glycan shifts the isotope envelope, so a peptide-averagine “most-abundant isotope” is frequently the wrong tooth, and the Byonic reference RT is the PSM scan RT (not the chromatographic apex). Both depress the centre-hit rate for the real control (26% vs 38–49% elsewhere) and for the unmatched set; the true glyco number is higher than the $\approx 3,800$ estimate but is not cleanly recoverable with a peptide base-peak model.
- Parameters held fixed: RT grace past elution end = 0.2 min; isolation-centre tolerance = 20 ppm; isolable-isotope floor = 5% relative abundance. Widening the RT grace or the ppm tolerance raises all three sets together and does not change the decoy-subtracted conclusion.

7. Reproduction

All under `analysis/decoy_features/`. Detector/scorer in `crates/flashlfq-core` (branch `decoy-features`).

- MS2 dump: `crates/flashlfq-core/examples/dump_ms2_precursors.rs` $\rightarrow$ `ms2_{10,65,2h}.tsv`.
- Score distributions + AUC: `plot_decoys.py`; ROC curves: `roc_curves.py`.
- Thresholds: `recall_threshold.py`, `precision_threshold.py`.
- Linkability: `ms2_linkability.py` ($\rightarrow$ `ms2_linkability.json`), `linkability_plot.py`, `net_estimate.py`.